

Mihal Dimo

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EDUCATION

University of Catania

Sep 2021 – 2025

Bsc in Computer Science

Relevant Modules: **Embedded systems, Algorithm & Data Structures, Operating Systems**

Brandenburgische Technische Universität Cottbus

Feb 2024 – Sep 2024

Bsc in Computer Science

Exchange Study Erasmus+

Relevant Modules: **Mathematical methods for Artificial Intelligence, Calculus, Software Security**

EXPERIENCE

Software Development Engineer Intern

Jan 2026 – Aug 2026

Amazon

Berlin, Germany

- Selected for an upcoming 7-month internship focused on cloud infrastructure and distributed systems.

Embedded Software Engineer

Mar 2024 – Present

KBDFans (Partner)

Changzhou, Jiangsu, China

- Technical Lead for **firmware/hardware** development (team of 5 C/C++ engineers) of the **Zellia Hall Effect** project, supporting 4+ **layout** with enhanced **modularity**.
- Streamlined **multi-layer PCBs** in **KiCad 8.0** with custom **Hall sensor footprints**; optimized for **precision** and flexible **switch configurations**.
- Boosted **ADC scan rate** from 1kHz to 106kHz (**+10500%**) using **multi-MCU** via **UART** communication and **Parallel ADC**, reducing key press and release latency by **90%**.
- Built **Open Source** cross-platform configuration tool in **Tauri/SvelteKit** with **real-time input tuning**, **0.005mm resolution**, and 8+ **programmable layers**.

PROJECTS

Zellia Firmware

- **Architected a distributed master-slave system** using 5 AT32 MCUs, decoupling **signal acquisition** from logic to achieve a **input latency** by <0.28ms.
- **Engineered a custom 7.5 Mbps UART protocol**, optimizing **bandwidth** to handle **high-frequency** per-key analog data synchronization without packet loss.
- **Reduced master CPU load by 90%** by shifting **real-time ADC normalization** and **distance calibration** to the slave node, ensuring **system stability** and **low-latency** input.
- **Developed a robust update mechanism**, allowing the master MCU to re-flash slave firmware dynamically, eliminating the need for device disassembly.

Zellia Control - Desktop Configuration Software

- Architected **cross-platform** App using **Tauri**, **SvelteKit**, and **Rust**, delivering **real-time keyboard configuration** with **0.005mm resolution control**.
- Implemented advanced **key modes**, supporting different configurations across 4 configurable layers with **profile import/export capabilities**.
- Integrated **responsive** design with **real-time keyboard visualization**, reducing eye strain.

osu! Chat Sentiment Analysis

- Built a **real-time sentiment analysis pipeline** for osu! chat using **Kafka**, **Elasticsearch**, and **Spark NLP**; processed **200,000+ messages/day** with **sub-second latency**.
- Achieved **86% classification accuracy**; used interactive **Kibana dashboards** for moderation insights.
- Improved **moderator efficiency by 50%** by automating **toxic content flagging** and reducing manual backlog recovery time.

Badge Weighting System V2

- Engineered a **ranking algorithm** for osu! players integrating **badge age** into performance metrics to improve fairness by **90%** in osu! tournaments in **C++**.
- Integrated **caching** and simplified **parsing complexity by 60%**, cutting **computation time by 25%** and reducing **date-related parsing errors by 100%**.

TECHNICAL SKILLS

Languages: C, C++, Rust, Python, TypeScript/JavaScript, Java, HTML/CSS.

Embedded & Hardware: ARM Cortex-M, CherryUSB, QMK keyboard firmware, KiCad (PCB design).

Web / Full-Stack: SvelteKit, Tauri (Rust), Node.js (Bun/Yarn/Npm/Pnpm), Tailwind CSS (HTML/CSS).

Data & DevOps: OpenCV, NumPy, Matplotlib, Apache Spark, Kafka, Elasticsearch, Kibana, Docker.

Tools: Git, VSCode, Bash, LaTeX, Linux.